Thermal Guardian

Team briefing & d-day reference

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Internal document — not for redistribution. Prototype; regulatory application in progress.

Thermal Guardian

Team briefing & d-day reference. Use this when a visitor asks something the presentation doesn't cover.

1. The Problem

> **19,000 people die each year** in bathtubs in Japan. **Over 90 % are 65 or older.** That's about **4× the annual traffic deaths.**

Why Japan specifically

- Long, deep hot baths (38–42 °C, 15–30 min) are the cultural norm
- Older Japanese homes have poorly insulated bathrooms and changing rooms → large cold-to-hot temperature swings
- Aging population (29 % are 65 +) → more people exposed to the highest risk

Three types of bathing risk

Category	What it is	Examples
🩺 **Physiological**	Body-state changes	Blood-pressure swings, heatstroke, cardiac stress
🏃 **Kinematic**	Motion / posture	Fainting, slipping, falling, silent drowning
🌡️ **Environmental**	Surroundings	Sudden temperature change ("heat shock")

Most accidents happen when **two or three of these stack** in the same minute.

Response window

Time under water	Outcome
3–4 min	Brain damage from oxygen deprivation begins
4–6 min	Death if no rescue
Every 1–2 min delay	Survival chance drops sharply

Implication: reaction alone isn't enough. We need **prevention before submersion.**

Sources

- Kanawaku Y. (2022). *Deaths in Bathtubs in Japan: Forensic and Clinical Implications.* J. Nippon Medical School 89(1), 33–39.
- Suzuki et al. (2017). *Bath-related deaths: preventive strategies.* J. General & Family Medicine.
- MHLW (2023). Bathtub drowning statistics.

2. Medical Background

Four main pathways lead to bath deaths. The system is designed around detecting all four.

2.1 Heat shock (ヒートショック)

What it is. Rapid blood-pressure swings triggered by a sudden temperature change — typically walking from a cold changing room into a hot bath or vice versa.

Mechanism.

1. Cold environment → blood vessels constrict → **BP spikes**
2. Hot water immersion → vessels dilate → **BP drops fast**
3. Elderly vessels can't adapt quickly → risk of stroke, heart attack, fainting

Our trigger. Temperature difference ≥ **15 °C** between changing room and bath area.

2.2 Orthostatic hypotension (起立性低血圧)

What it is. Blood pools in the lower body during hot immersion. When the person stands up too fast, the heart can't recover blood pressure quickly enough → brain hypoxia → fainting.

Mechanism in one picture.

```
Hot water    →  Vessels max-dilated, blood pools in legs
Stand up     →  Gravity pulls blood down
Old heart    →  Can't raise HR fast enough to push blood up
Brain        →  Hypoxia → FAINT
Fainting     →  Head under water → silent drowning
```

Why elderly. Aging reduces autonomic reflex speed; vessels are stiffer.

2.3 Heatstroke / hyperthermia (熱中症)

What it is. Core body temperature rises faster than the body can dissipate heat. Happens when the bath is too hot, too long, or ventilation is poor.

Our thresholds (escalating voice reminders):

Bath time	System says
15 min	"You've been bathing 15 minutes — check how you feel."
20 min	"20 minutes elapsed — consider finishing soon."
25 min	⚠️ "Warning: long bath time. Exit within 5 minutes."
30 min	🚨 "Danger: 30 minutes elapsed. Exit immediately."

2.4 Cardiovascular events

What. Sudden cardiac arrest, arrhythmia, or myocardial infarction triggered by thermal stress.

Risk factors. Existing heart disease, hypertension, atherosclerosis — all common in 65 +.

Detection. Abnormal HR / HR variability patterns + context (is the user in BATHING state? just stepped out into EXIT_BATHTUB?).

2.5 Common outcome: drowning

All four pathways above can end the same way:

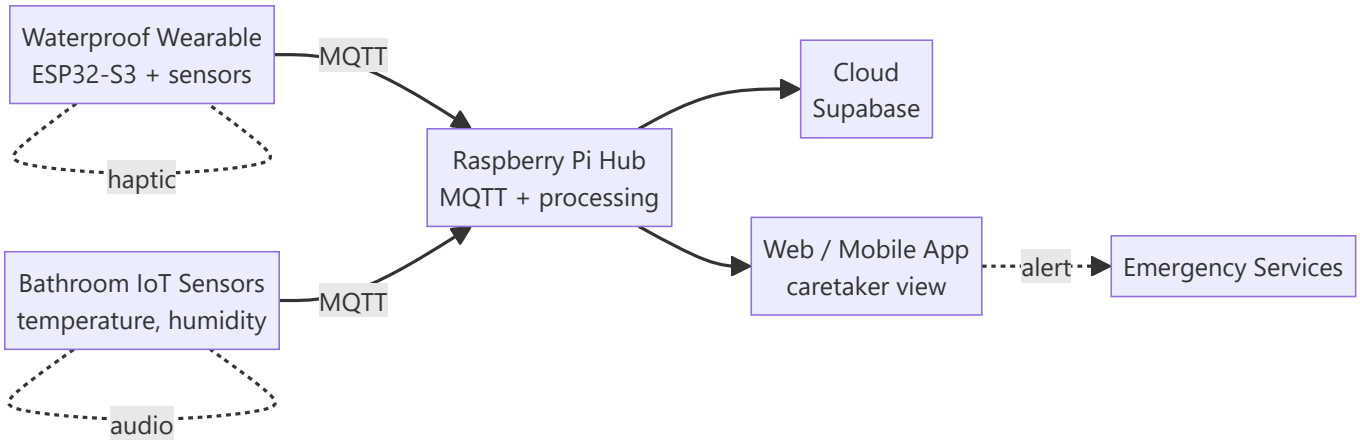
Loss of consciousness → submerged face → silent drowning. ("Silent" because elderly tend to slide quietly under — no struggle, no call.)

Sources

- Satoh et al. (2015). *Regressive changes of myocardial sleeves in sudden bathtub death.*
- Steward et al. (2023). *Age & mitigation strategies during hot water immersion — orthostatic intolerance.* Experimental Physiology.
- Narisada et al. (2005). *Bathing-related accidents requiring ambulance dispatches.*

3. How the System Works

3.1 Overview



Three layers:

1. **Wearable + room sensors** — collect raw signals
2. **Raspberry Pi hub** — runs the algorithm locally (no cloud round-trip for alerts)
3. **Cloud + apps** — persistence, caretaker dashboard, emergency escalation

Why local MQTT first? Latency. A cloud round-trip can cost 500 ms+ per message. In an emergency, alerts go wearable → Pi → user's haptics in **< 50 ms**.

3.2 What the system continuously checks

Signal	Sensor	Why
Cardiovascular (HR, SpO_2)	MAX30101 + MAX32664 (PPG)	Faint / cardiac event detection
Movement + posture	ISM330DHCX (9-axis IMU)	Fall / slip / "stood up too fast"
Body temperature	MCP9808 / MAX30205	Heatstroke onset
Environment temp + humidity	SHT40 (MEMS)	Heat shock (Δ between rooms)

3.3 Wearable hardware at a glance

- **MCU:** Seeed Studio XIAO ESP32-S3 (Wi-Fi, Bluetooth, dual-core)
- **Power:** 3.7 V 500 mAh LiPo + TP4056 charger
- **Output:** Vibration motor (haptic). Audio output is on the bathroom IoT unit, not the wearable.
- **Waterproofing:** IPX7 target — conformal coating, silicone grease, PTFE breathable membrane, RTV sealing
- **Form factor:** Armband; waterproof USB-C for charging

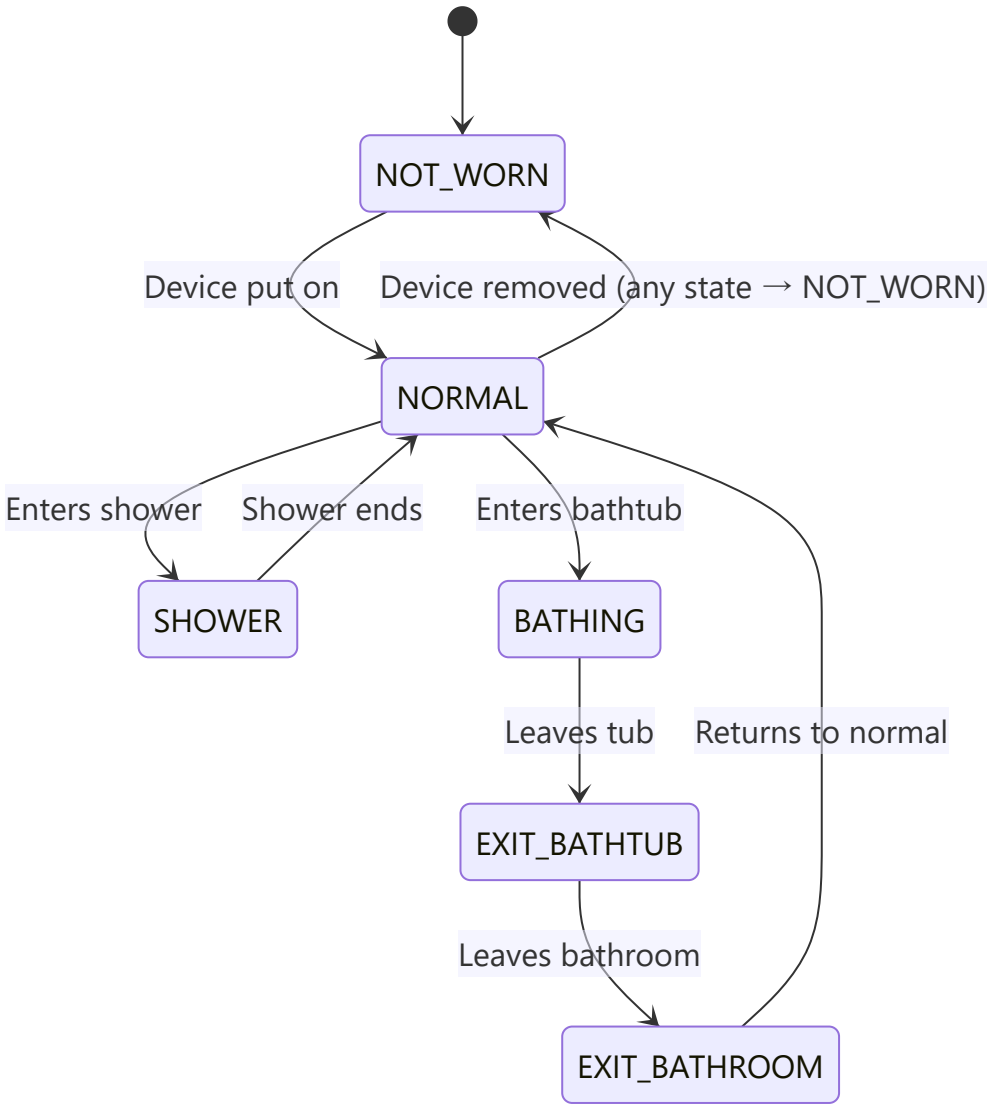
3.4 Three things that make Thermal Guardian different

1. 🧿 **Multi-modal sensing** — body + motion + environment in one system, not one alone
2. 🧠 **Smart safety algorithm** — different risk checks per bathing stage
3. 🔒 **Privacy-first** — no cameras ever; alert logic runs locally on the Pi

4. Clinical Logic

4.1 Six bathing states (state machine)

The system tracks which state the user is in. A rule that makes sense during BATHING may be noise during NORMAL. Entering and exiting the bathroom are derived from humidity. **NOT_WORN** is a global override reached from any state on device removal; on re-wear, the system returns to the previous state.



States per the team's Medical Rules table, 2026-02-13 meeting materials.

4.2 Ten medical rules (M-001 to M-010)

Rule catalog verbatim from the team's Medical Rules table (2026-02-13 meeting). Each rule fires only in its active state(s). Action semantics: **WARNING** = haptic on wearable + voice on the bathroom IoT unit + caretaker app notice; **EMERGENCY** = immediate escalation to emergency services.

#	Name	Trigger	Action	Active state(s)
M-001	Fall + syncope escalation	Impact event → 30 s assessment window; within 30 s, if HR < 40 sustained 10 s OR (HR_max_10s − HR_avg_10s) > 20	WARNING → EMERGENCY	All states

#	Name	Trigger	Action	Active state(s)
M-002	HR high	HR > 120	WARNING	BATHING
M-003	HR rise 30	HR_avg_10s − HR_IMMERSION_REF > 30	WARNING	BATHING
M-004	HR rise 35	Sustained (HR_avg_10s − HR_IMMERSION_REF) > 35	WARNING	BATHING
M-005	Heat rate	slope(Tcore_est) > 0.13 °C/min	WARNING	NORMAL, SHOWER, BATHING, EXITBATH
M-006	Tcore 38.5	Tcore_est ≥ 38.5 °C	WARNING	NORMAL, SHOWER, BATHING, EXITBATH
M-007	Tcore 37.5	Tcore_est ≥ 37.5 °C	WARNING	NORMAL, SHOWER, BATHING, EXITBATH
M-008	Room cold	Non-elderly: T_changing < 15 °C; elderly: T_changing < 20 °C	WARNING	NORMAL, EXIT_BATHTUB
M-009	10-min bathing reminder	time_in_BATHING ≥ 10 min AND bathing_10min_warned == false (fires once)	WARNING	BATHING
M-010	HRV low *(currently disabled)*	After 60 s in BATHING, HRV_pNN50 < 0.5 × HRV_REF_pNN50 sustained 30 s	WARNING	BATHING

> ⚠ All HRV processing is currently disabled until reliable HRV extraction is in place. M-010 is listed for completeness but does not fire in the current system.

4.3 Escalation ladder

```
Rule fires
  ↓
1. User haptics + voice prompt
  ↓ (no response)
2. Caretaker / family notified via app
  ↓ (no response)
3. Emergency services + nearest hospital contacted
```

4.4 What is NOT validated yet

- 🔴 **Not FDA / CE / PMDA certified.** Research prototype. Regulatory application in progress.
- 🔴 **Clinical trials pending.** Accuracy figures in this doc are from internal testing, not peer-reviewed clinical validation.
- 🔴 **The advertised accuracy is the goal, not a proven outcome.** Next phase: supervised data collection with older adults → clinical validation.

If asked bluntly — "is this approved for medical use?" — the correct answer is:

> **"It is a research prototype. Regulatory application is in progress. We are not yet cleared for clinical deployment."**

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5. Cost

5.1 iCAN-approved budget (per the pitch deck)

Category	Amount (JPY)
Device components	¥ 34,694
Materials (PLA, misc)	¥ 20,198
Travel (flight + accommodation)	¥ 40,000
Total approved	**¥ 95,892**

5.2 Actual per-unit wearable BOM (from inventory)

Key components only — full list in `Thermal Guardian Inventory.xlsx`.

Component	Unit cost (JPY)
Seeed XIAO ESP32-S3 (MCU)	3,759
ISM330DHCX IMU module	12,555
MAX30101 + MAX32664 PPG	8,238
SHT40 temp/humidity	1,200
MAX30205 body-temp chip	1,612
MAX98357 audio amp	749
Vibration motor	803
LiPo 3.7 V 500 mAh	1,199
TP4056 charger	750
Enclosure + waterproofing (amortized)	~ 3,000
Wearable sub-total (rough)	**~ ¥ 34,000**

Plus Raspberry Pi 4 hub (~ ¥ 10,000) shared across a household.

5.3 Realistic deployment cost per household

- 1 wearable per elderly user: **~ ¥ 34,000**
- 1 Pi + bathroom sensors per home: **~ ¥ 15,000**
- **Total household hardware: ~ ¥ 50,000**

Ongoing cloud costs (Supabase free tier for pilot; scaling tier ≈ ¥ 3,000 / month per 100 users).

6. Competitive Landscape

This is the section most likely to be pressed by industry judges.

6.1 Japanese market — direct competitors

Product	Tech	Strengths	Weaknesses	Approx cost
Meta 浴槽見守り (Metatechno + Medical Project Inc.)	Pressure-mat sensor on bathing bench (1 Pa, 0.1–20 Hz) — breathing / pulse / body motion	Co-developed with Medical Project Inc. Launched as prototype at Care Technology Tokyo 2025.	Prototype only. Cannot detect falls. Bench/tub-specific.	Not disclosed (prototype)
Bath Safe (Medical Project Inc.)	Air-structure mat on tub floor (suction cups) + optional auto-drain	Automatic drainage on anomaly. Institutional track record (施設向け).	Tub-only. Cannot detect falls outside tub. No body monitoring. Vendor disclaimer: "not for life-saving."	Not disclosed (B2B institutional)
Kyokko System (旭光電機)	Optical beam sensor + alarm; METI robot-care program	Continuous beam detection	Only triggers after submersion. False alarms common. No pre-event prevention.	Not disclosed (B2B institutional)
Shinsei HCU	Ring PPG + SpO_2 (IP68) + bathroom unit (human PIR + radar + sonic) + speaker/mic + auto-drain	**Closest architectural match to ours.** Multi-modal, privacy-first, partially preventive.	Still 開発中 as of 2026. No ship date, no field data, no PMDA class.	Not disclosed (in development)
JVC Kenwood 浴室あんしん安全システム	Ceiling-mounted ultrasonic; head-position tracking	3 s detection, 18 s external notify. Targeting house-makers, サ高住, security companies for B2B partnerships.	Ceiling-only, no wearable, no body vitals, no environment sensing. Prototype since 2020, still not commercialized.	Not disclosed (prototype, B2B only)
Noritz おふろの見まもり / 安心入浴サポート	PIR + water-level sensor in gas-water-heater remote; わかすアプリ smartphone notification	Integrated into existing plumbing. Nationwide plumber distribution.	Occupancy + timer only. No vitals, no fall detection, no heat-shock model. Requires a Noritz heater.	≈ ¥426,000 for heater (GT-C2462 series; monitor included in remote)

6.2 International — wearable fall / health

Product	Tech	Strengths	Weaknesses for *bathing* use	Approx cost
Apple Watch Series 10 / Ultra 2	PPG, IMU fall detection, SpO_2, temperature	Improved fall-detection SIP (S10); mass market	Apple explicitly warns that soap, shampoo, and bath products **degrade the water seals** — bath use voids the water-resistance guarantee. No bath-context algorithm. No environment sensors. Reactive only.	¥107k–¥161k (one-time, JP retail)
Samsung Galaxy Watch Ultra (2025)	IP68 + 50 m freshwater; PPG + BIA	Vendor-certified for shower use	**Can survive bath contact, but has no bath-state algorithm, no environment sensors, no heat-shock / immersion model.**	~US$600 (≈ ¥90k)
Fitbit Sense 2 / Charge 6	HR, skin temp, EDA	Sleep / stress tracking	No fall detection in bath context. No immersion model. No caregiver escalation.	¥24k–¥38k (one-time)
MEDIROM REMONY MOTHER Bracelet	PPG HR + skin temp + 3-axis IMU + impact fall + SOS; self-charging (thermoelectric + solar); 5 ATM	Self-charging uniquely solves wear-compliance; real multi-sensor wearable	Workplace-primary (warehouse heat-stroke), eldercare secondary. No bath-context model, no SpO_2, no environmental pressure/humidity, no PMDA class.	On request (B2B, not publicly disclosed)

6.3 Japanese emergency-response services (adjacent, not direct)

Service	Installed base	What it does	Why it's not a substitute	Approx cost
セコム (Secom) Home Security — マイドクター pendant	~2.54 M contracts (2023-09)	Push-button emergency call; splashproof outside the tub	**Reactive** — user must still be conscious to press the button. Vendor says pendant not for inside-tub use.	親の見守りプラン rental ¥5,060/mo + install ¥48,400 + deposit ¥20,000 (or buyout ¥219,890 + ¥3,410/mo)
ALSOK みまもりサポート	~1.78 M contracts (2022-05); No. 1 in municipal deployments	Emergency button + elder-check	Same limitation. Pendant not recommended in the bathroom at all.	Rental ¥2,750/mo + install ¥13,200 (5-yr contract); entry plans from ¥990/mo
MCI waterproof call-pendant services	n/a	Waterproof pendant with call button	Requires user action during emergency.	Varies by provider

6.4 Academic / research — related work

- **Wi-Fi CSI bathing monitoring** (various universities) — detects occupancy + motion via Wi-Fi signals. No body monitoring. Infrastructure-dependent.
- **Ambient radar bath monitoring** — similar concept; no body-state data.
- **Camera-based systems** — rejected on privacy grounds (our design specifically avoids cameras).

- **Tokyo Gas × 日本気象協会 ヒートショック予報** — weather-based risk advisory on tenki.jp (Oct–Mar, 2017–present). Not a device — a complementary ambient signal ThermalGuardian can consume.

6.5 Our positioning in one sentence

> Thermal Guardian is the only Japan-market system that combines **body state + motion + environment** on a **privacy-preserving wearable**, backed by a **bath-context state machine and ten numbered medical rules (M-001 – M-010)**, with a **working field-testable pipeline** — while comparable architectures (Shinsei HCU) remain in development.

Positioning summary

Axis	Existing	Thermal Guardian
Monitoring target	Tub, room, or body (one)	All three at once
Alert timing	After the accident	Before the accident
Privacy	Cameras / room beams	Wearable only, no cameras
Deployment	Fixed bathroom installation	Wearable — works at home, onsen, hotel
Bath-context algorithm	None (or reactive only)	6-state FSM + M-001 – M-010
Pipeline maturity	Prototype / development	30+ commits, 250+ passing tests

ThermalGuardian approximate cost (for direct comparison)

- **Per-unit wearable BOM ≈ ¥34,000** (one-time, see §5)
- **Household deployment ≈ ¥50,000** (one-time; wearable + IoT unit + Pi hub)
- **No monthly fee** in the current architecture (local-MQTT-first, no mandatory cloud subscription)
- Cheapest direct-competitor monthly service (ALSOK light plan ¥990/mo) reaches ThermalGuardian's household cost in **~50 months**; Secom's standard plan reaches it in **~10 months**. Apple Watch S10 alone (¥107k+) is already 2× our household deployment with no bath-context algorithm.

Regulatory note

All Japanese bathing-safety competitors surfaced sit in the unregulated "lifestyle / information device" category (no PMDA 医療機器 class). ThermalGuardian's "research prototype, regulatory application in progress" phrasing is **consistent with category norms, not behind them**.

Go-to-market note

介護保険 福祉用具貸与 (public long-term-care rental) does **not** cover bathroom monitoring sensors as of 2025 — the 13-item covered list includes dementia-wandering sensors and bed-leaving sensors only. Every competitor including ThermalGuardian must sell direct-to-consumer or B2B to facilities; **public reimbursement is not a current path**.

Market sizing

Japan elderly-monitoring + emergency-call market: **¥22.7 B (2025 forecast, Seed Planning)**. Care-DX market: **¥54.76 B (FY2024, Yano Research)**. Bath-specific sub-market has no public sizing.

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7. Glossary

7.1 Medical terms

JP Japanese	GB English	Meaning (one line)
入浴 (にゅうよく)	Bathing	The act of taking a bath
入浴事故 (にゅうよくじこ)	Bathing accident	Death or injury during bathing
ヒートショック	Heat shock	Rapid BP swing from temperature change
起立性低血圧 (きりつせいていけつあつ)	Orthostatic hypotension	BP drop when standing up
血管迷走神経性失神 (けっかんめいそうしんけいせいしっしん)	Vasovagal syncope	Fainting from autonomic reflex
熱中症 (ねっちゅうしょう)	Heatstroke	Body overheating beyond dissipation capacity
高体温症 (こうたいおんしょう)	Hyperthermia	Abnormally high core body temperature
心筋梗塞 (しんきんこうそく)	Myocardial infarction	Heart attack
不整脈 (ふせいみゃく)	Arrhythmia	Irregular heart rhythm
意識障害 (いしきしょうがい)	Loss of consciousness	Fainting / unresponsive
血圧 (けつあつ)	Blood pressure	Force of blood against vessel walls
心拍数 (しんぱくすう)	Heart rate	Beats per minute
血管 (けっかん)	Blood vessel	Artery / vein
血管拡張 (けっかんかくちょう)	Vasodilation	Vessels widening (e.g. in heat)
血管収縮 (けっかんしゅうしゅく)	Vasoconstriction	Vessels narrowing (e.g. in cold)
深部体温 (しんぶたいおん)	Core body temperature	Internal organ temperature
皮膚温 (ひふおん)	Skin temperature	Surface temperature
溺水 (できすい) / 溺れ (おぼれ)	Drowning	Death / harm by submersion
声なき溺れ (こえなきおぼれ)	Silent drowning	Submersion with no calls or struggle
自律神経 (じりつしんけい)	Autonomic nervous system	Involuntary nervous control
高齢者 (こうれいしゃ)	Elderly	65 + target demographic
介護者 (かいごしゃ)	Caretaker	Family member or pro caregiver
脱衣所 (だついじょ)	Changing room	Pre-bath room, often unheated

7.2 Technical terms

Term	What it means
MEMS	Micro-Electro-Mechanical Systems — tiny sensors etched into silicon
IMU	Inertial Measurement Unit — accelerometer + gyroscope (± magnetometer)
9-axis	3-axis accelerometer + 3-axis gyroscope + 3-axis magnetometer
PPG	Photoplethysmography — optical heart-rate sensing
SpO_2	Blood oxygen saturation percentage
HRV	Heart-rate variability — beat-to-beat timing changes
MCU	Microcontroller — the wearable's "brain" (ESP32-S3)
ESP32-S3	Dual-core Wi-Fi + BT MCU by Espressif
MQTT	Lightweight pub/sub messaging protocol for IoT
FreeRTOS	Real-time operating system for embedded devices
Mosquitto	The MQTT broker software running on the Pi
I^2C / SPI	Serial communication buses for connecting sensors to an MCU
FSM	Finite State Machine — logic that moves between defined states
Markov chain (マルコフ連鎖)	A state-transition model where next state depends on current state
IPX7 (防水等級 IPX7)	Waterproof rating: 30 min immersion in 1 m water
Sampling rate	How many times per second a sensor is read (ours: 100 Hz raw, 1 Hz published)
Haptic feedback	Vibration-based notification
Raspberry Pi	Credit-card-sized Linux computer used as the hub
Supabase	Cloud database (PostgreSQL) — stores history
ngrok	Tunneling service for remote access during prototyping
PCB	Printed Circuit Board — the wearable's electronics carrier
Conformal coating	Thin protective layer (silicone) over circuits for water resistance
PTFE membrane	Waterproof but air-permeable fabric — lets pressure equalize

Emergency short-answer card (if pressed for a 30-second reply)

> **"Thermal Guardian is a wearable that prevents bathing accidents in Japan's elderly — where 19,000 people die every year. We detect heat shock, heatstroke, fainting, and cardiac events by combining body, motion, and environment sensors. When risk is rising, we warn the user before they black out — and if they do, we alert caretakers and emergency services in seconds. No cameras. Fully private. Research prototype, regulatory application in progress."**

Document ends. All terminology should match the poster and booth scripts. Last updated 2026-04-19.